

## Third Terminal Examination - 2082

Class: XI

Subject: Mathematics (0071)

Full Marks: 75

Time: 3 hrs.

SET 'A'

Pass Marks: 30

*Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.*

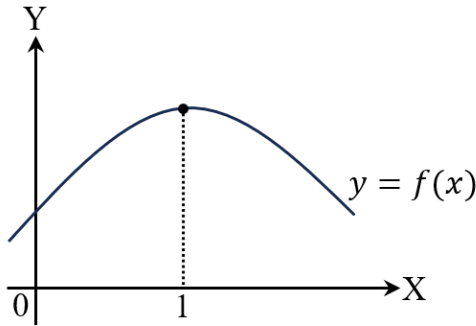
**Attempt all the questions.**

### Group 'A'

**Rewrite the best alternative of each question in your answer sheet. [11 × 1 = 11]**

- The converse of the statement " $p \Rightarrow q$ " is  
 a)  $\sim p \Rightarrow \sim q$       b)  $q \Rightarrow p$       c)  $\sim q \Rightarrow \sim p$       d)  $p \wedge \sim q$
- A function  $f: \mathbb{R} \rightarrow [0, \infty)$  defined by  $f(x) = x^2$  is  
 a) onto function      b) one-one function  
 c) bijective function      d) Neither
- If  $A = \begin{pmatrix} 0 & 2x+1 \\ x-4 & 0 \end{pmatrix}$  is skew-symmetric matrix then  $x =$   
 a)  $-1$       b)  $2$       c)  $3$       d)  $1$
- The equation  $x^2 + y^2 + 2gx + 2fy + 1 = 0$  represents a pair of straight lines if  
 a)  $f^2 + g^2 = 1$       b)  $f^2 - g^2 = 1$   
 c)  $f^2 + g^2 = 0$       d)  $f^2 - g^2 = 0$
- If the sum of the roots of the equation  $qx^2 + 2x + 3q = 0$  is equal to their product, then the value of  $q$  is  
 a)  $\frac{3}{2}$       b)  $-3$       c)  $\frac{-2}{3}$       d)  $3$
- Let  $B_1$  and  $B_2$  be two bisectors of the angle between the lines  $AB$  and  $CD$ . If  $\theta$  be the angle between  $B_1$  and  $AB$ , then  $B_1$  will be acute angle bisector if  
 a)  $|\tan \theta| > 1$       b)  $|\tan \theta| < 1$   
 c)  $|\tan \theta| = \infty$       d)  $\tan \theta = 1$

7. The direction cosines of the x-axis are  
 a) 0, 0, 1                      b) 0, 1, 0                      c) 0, 1, 1                      d) 1, 0, 0
8.  $\lim_{x \rightarrow \infty} \frac{\sin x}{x} =$   
 a) 1                      b) -1                      c) 0                      d) does not exist.
9. If the function  $f(x) = \begin{cases} \frac{\tan px}{x} \end{cases}$  is continuous at  $x = 0$ , then  $p =$   
 a) -6                      b)  $\frac{1}{6}$                       c) 6                      d)  $-\frac{1}{6}$
10. If  $y = \ln(\ln e^x)$  then  $\frac{dy}{dx} =$   
 a)  $\frac{1}{e^x \ln(e^x)}$                       b)  $\frac{1}{x}$                       c)  $\frac{e^x}{\ln(e^x)}$                       d)  $e^x$
11. Which of the following is true for the adjoining figure?



- a)  $f'(x) > 0$  for  $(-\infty, 1)$ ,  $f'(x) < 0$  for  $(1, \infty)$  and  $f''(x) < 0$  for all  $x$ .  
 b)  $f'(x) > 0$  for  $(-\infty, 1)$ ,  $f'(x) < 0$  for  $(1, \infty)$  and  $f''(x) > 0$  for all  $x$ .  
 c)  $f'(x) > 0$  for  $(-\infty, 1)$ , and  $f''(x) > 0$  for all  $x$ .  
 d)  $f'(x) > 0$  for  $(1, \infty)$ , and  $f''(x) < 0$ .

### Group 'B'

#### Short Answer Questions.

[8 × 5 = 40]

12. a) Construct the truth table for (i)  $\sim (p \wedge q)$  and (ii)  $\sim p \vee \sim q$ . Are they logically equivalent? Explain. [1+1+1]  
 b) For any  $a, b \in \mathbb{R}$ , prove that:  $|a + b| \leq |a| + |b|$  [2]

13. a) If  $x = \log_{2a} a$ ,  $y = \log_{3a} 2a$  and  $z = \log_{4a} 3a$ , prove that:  
 $xyz + 1 = 2yz$ . [3]
- b) Let  $\mathbb{R}$  be the set of real numbers and the function  $f: \mathbb{R} \rightarrow \mathbb{R}$  is defined by  $f(x) = 2x - 3$  for all  $x \in \mathbb{R}$ . Does  $f^{-1}$  exist? [2]
14. a) Find inverse of the matrix  $A = \begin{pmatrix} -1 & 1 & 0 \\ 3 & -2 & 1 \\ 2 & -2 & -1 \end{pmatrix}$  if exists. [3]
- b) Define conjugate of a complex number. If  $z = \frac{(2, 3)}{(3, -2)}$  then find  $\bar{z}$  in the form of  $a + ib$ . [1+1]
15. a) If  $m$  and  $n$  are the lengths of perpendiculars from origin upon the straight lines  $x \sec \alpha + y \operatorname{cosec} \alpha - b = 0$  and  $x \cos \alpha - y \sin \alpha = b \cos 2\alpha$  respectively, prove that:  $4m^2 + n^2 = b^2$ . [3]
- b) Find the value(s) of  $m$  so that equations  $x^2 - mx - 21 = 0$  and  $x^2 - 3mx + 35 = 0$  have a common root. [2]
16. a) Solve:  $z^2 = -15 - 8i$  [3]
- b) Find the ratio in which the line joining the points  $(1, 2, 3)$  and  $(4, -4, 9)$  is divided by the  $zx$  -plane. [2]
17. a) Evaluate:  $\lim_{x \rightarrow \theta} \frac{x \tan \theta - \theta \tan x}{x - \theta}$  [3]
- b) Define direction ratio of a line. Find the direction cosines of a line whose direction ratios are 1, 2, 2. [1+1]
18. a) Find, from first principle, the derivative of  $f(x) = \sqrt{\sin 2x}$ . [3]
- b) If  $\cos(x - y) = \ln(x - y)$ , then prove that:  $\frac{dy}{dx} = 1$ . [2]
19. a) Write the sufficient condition for a function  $f(x)$  to have minimum value at  $x = a$ . Prove that the function  $f(x) = x^3 - 6x^2 + 12x + 5$  has neither maxima nor minima. [1+2]
- b) Find  $\frac{dy}{dx}$ , when  $y = \cos^{-1}(2x^2 - 1)$ . [2]

### Group 'C'

#### Long Answer Questions.

[3 × 8 = 24]

20. a) Let  $A, B, C$  be three non-empty sets. Prove that:  
 $(A \cup B) - C = (A - C) \cup (B - C)$  [2]

b) Find domain and range of the real valued function  $f(x) = \frac{1}{\sqrt{x-3}}$ . [2]

c) If  $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$  and  $B = \begin{pmatrix} -1 & 2 \\ 5 & -4 \end{pmatrix}$ , then show that:  $(AB)^T = B^T \cdot A^T$ . [2]

d) Prove that: 
$$\begin{vmatrix} 1+x_1 & x_2 & x_3 \\ x_1 & 1+x_2 & x_3 \\ x_1 & x_2 & 1+x_3 \end{vmatrix} = 1 + x_1 + x_2 + x_3$$

21. a) Prove that the straight lines joining the origin to the point of intersection of the line  $lx + my + n = 0$  and the circle  $x^2 + y^2 = a^2$  are at right angles if  $2n^2 = a^2(l^2 + m^2)$ . [3]

b) Let two lines be  $2x + 3y - 5 = 0$  and  $3x + 2y + 7 = 0$ .

i) Find the angle bisectors between them.

ii) Identify the bisectors containing origin. [1+1]

c) The sum of the roots of a quadratic equation is 2 and the sum of their squares is 20. Find the equation. [2]

d) Find the distance between two parallel lines  $\frac{x}{\sec \alpha} + \frac{y}{\csc \alpha} - 4 = 0$  and  $\frac{x}{\sec \alpha} + \frac{y}{\csc \alpha} + 5 = 0$ . [1]

22. a) Let  $f(x)$  be a function representing a curve. What does  $f'(x)$  represent geometrically? [1]

b) For the function  $f(x) = 2x^3 - 3x^2 - 12x + 14$ , find

i) slope of tangent to the curve represented by above function at  $(1, 0)$ .

ii) global minimum value in  $[-1, 3]$ .

iii) point of inflection. [1+1+1]

c) Evaluate:  $\lim_{x \rightarrow 16} \frac{x-16}{\sqrt{x}-4}$  [1]

d) A function  $f(x)$  is defined as,

$$f(x) = \begin{cases} 2x^2 - 5 & \text{for } x > 2 \\ 5 & \text{for } x = 2 \\ 7 - x^2 & \text{for } x < 2 \end{cases}$$

Show that  $f(x)$  has removable discontinuity at  $x = 2$ . Also, redefine  $f(x)$  to make it continuous. [2+1]

**☞ - The End - ☞**